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Subject: Proposal: Housing Unaffordability and Displacement Risk in Charlotte

This project will analyze the main factors influencing housing unaffordability in Charlotte and identify the neighborhoods considered at highest risk of being displaced. We will then continue to evaluate whether historically redlined neighborhoods are represented disproportionately within high-risk areas, providing evidence of long-term impacts.

Our data sources will integrate two primary datasets; Charlotte Area Quality of Life Dataset (QoL) and the Mapping Inequality dataset (HOLC or redlining). We will merge Neighborhood Planning Area (NPA) data from the Quality of Life dataset, which includes variables such as housing, socioeconomic status, demographics, and other infrastructure variables, along with the HOLC redlining grades from the Mapping Inequality dataset.

Model one, our regression model, will focus on housing unaffordability. Our target for this model is to predict the housing cost burden. Key features will consist of household income and employment, market conditions, neighborhood conditions, and demographic makeup. The goal is to use a multiple linear regression model and find a better model if possible. With the model outputs, we will be then able to identify and interpret the primary variables and influences of increased housing costs.

Model two, the classification model, will focus on the binary displacement risk, using existing vulnerability measurements. These features will include socioeconomic vulnerability status, like poverty, education, homeownership, along with housing pressure, and neighborhood change variables. The goal is to use a logistical regression approach for the model, but this is subject to change. The output will show predicted probability of displacement risk for specific neighborhoods, which can be possibly tied back to the redlining of neighborhoods.

We will then flag neighborhoods predicted at high-risk displacement risks and merge them with HOLC grades to compare the two. This will help evaluate whether historically redlined neighborhoods face higher risk of displacement today.